Effects of Integrating an AI Learning Companion into a Self-Regulated Learning 4L Framework

on Elementary Mathematics Learning

A quasi-experimental study on TALPer-supported SRL in fifth grade

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Full Paper · 15 min + 5 min Q&A

TALPer improves learning, but regulation explains who benefits

A 15-minute storyline: problem -> intervention -> evidence -> implication

TALPer improves mathematics learning, but its strongest lesson is differentiated: students benefit when they regulate, question, monitor, and reflect with AI.

Problem

GenAI may scaffold or create dependence

Intervention

TALPer embedded in SRL 4L

Evidence

Outcome, process, and ENA networks

Takeaway

AI access is not enough

The GenAI promise comes with a dependence risk

Structured AI feedback can scaffold SRL, but passive use may reduce learner agency

Scaffold

Real-time feedback
Step-by-step guidance
Planning, monitoring, and reflection
Differentiated support

(Chiu, 2024; Wu et al., 2023)

Dependence

Passive answer-seeking
Procedural shortcutting
Reduced learner agency
Ability-level differences

(Jin et al., 2023; Lee & Low, 2024)

Grounded in Zimmerman's (2002) SRL cycle: forethought -> performance -> self-reflection

Research gap: whether this scaffold-dependence tension differs across learner ability levels remains underexplored.

Zimmerman's SRL Cycle x Ho's 4L Framework

Forethought -> Performance -> Self-Reflection

Zimmerman (2002) SRL Cycle

Forethought
Goal setting - Prior-knowledge activation

Performance
Monitoring - Strategy - Peer adjustment

Self-Reflection
Error noticing - Conceptual consolidation

Ho (2014) SRL 4L Framework

Self-learning
Goal setting / prior activation

<-> Forethought

Co-learning
Monitoring & scaffolded feedback

<-> Performance

Mutual learning
Cross group strategy sharing

<-> Performance

Teacher-guided
Misconception fix & integration

<-> Self Reflection

The 4L phases map onto Zimmerman's cycle. We apply this mapping to design our SRL coding scheme and ENA dimensions.

TALPer is embedded inside the SRL 4L learning cycle

Self-learning -> co-learning -> mutual learning -> teacher-guided consolidation

Based on Ho's (2014) SRL 4L framework, grounded in Zimmerman's (2002) SRL cycle

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Original TALPer interface from the source deck

Student asks a mathematical question

TALPer responds through guided reasoning

Interaction logs become classroom evidence

This is the concrete system view behind the SRL 4L integration.

TALPer: an AI learning companion on TALP for elementary math problem solving (Kuo et al., 2026)

The study follows three evidence layers

Outcome, process, and cognitive network structure

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RQ1 · Outcome

Does TALPer + SRL 4L improve learning outcomes compared with SRL 4L alone?

RQ2 · Process

Do high- and low-achieving students differ in SRL strategy use when TALPer is available?

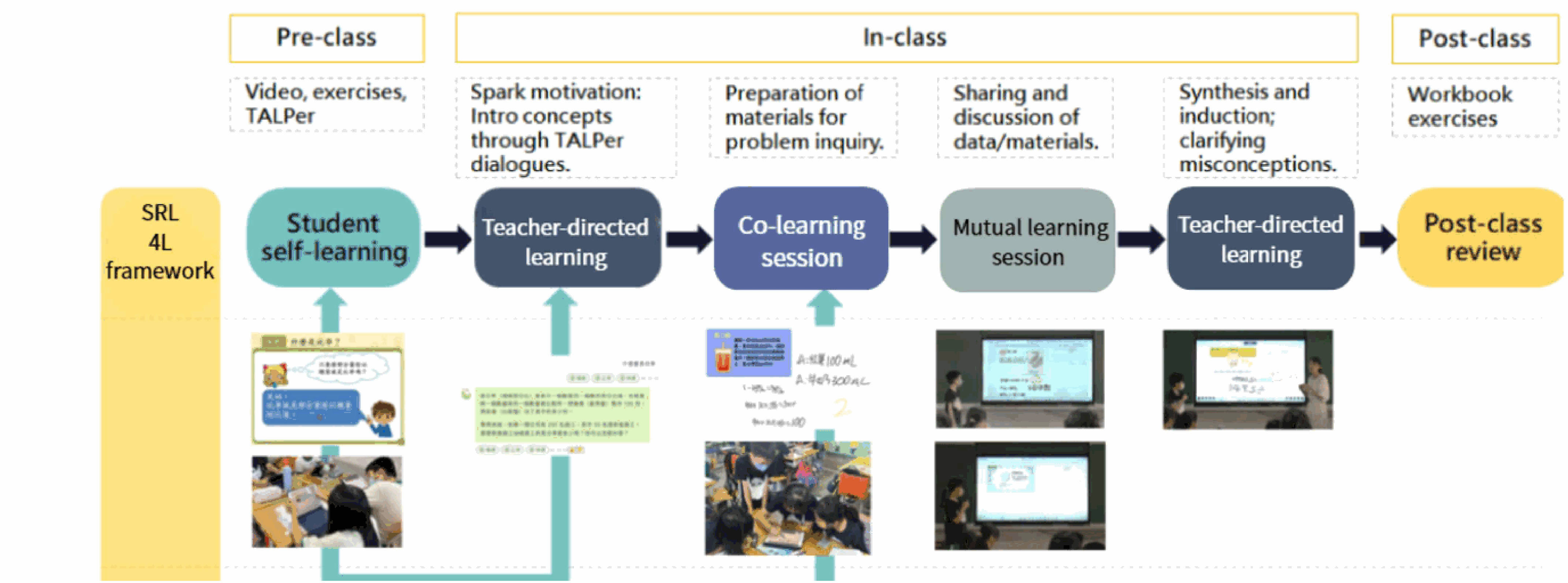
RQ3 · Structure

Do their cognitive network structures differ in ENA representations?

The comparison isolates TALPer within the same 4L structure

44 fifth-grade students, 5 lessons on Ratios and Percentages, 2-week intervention

Original SRL 4L teaching process from the source deck



Pre-class TALPer self-learning

In-class 4L discussion sequence

Post-class review and exercises

SRL Coding Framework and ENA Dimensions

Table 1 - Metacognition x Cognition - 8 codes

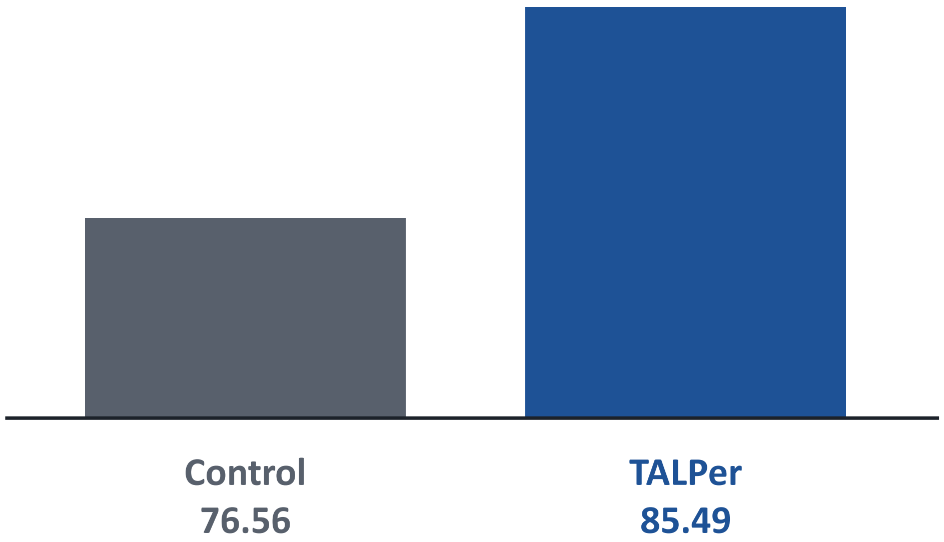
Main category	Sub-category	Code	Explanation
M Metacognition	Setting Goals	M.SG	Proactively sets or adjusts learning objectives or sequences.
	Selecting Strategies	M.SS	Selects or modifies learning strategies, modalities, or task forms.
	Monitoring & Evaluation	M.ME	Real-time awareness and evaluation of conceptual understanding.
	Reflection	M.R	Reviews problem solving to identify errors or revise strategies.
C Cognition	Conceptual Und. Ask	C.CUA	Asks questions about concepts, principles, meanings, or applications.
	Conceptual Und. Explain	C.CUE	Articulates or constructs conceptual understanding.
	Procedural Know. Ask	C.PKA	Asks about computational methods or problem solving steps.
	Procedural Know. Execute	C.PKE	Performs actual computations or formulates equations / answers.

ENA maps code co-occurrences into a 2D space -- X axis: relative cognitive tendencies; Y axis: structural complexity / diversity of the cognitive network.

TALPer-supported 4L produced higher adjusted post-test scores

After controlling for pre-test performance

Adjusted post-test means



ANCOVA result

$F(1,41) = 4.71$

$p = .036$

eta squared = .114

Takeaway: TALPer adds measurable value beyond SRL 4L instruction alone.

Pre-post gains: Control +14.45; TALPer +26.45 (both $p < .001$)

High- and low-achieving students used TALPer differently

The major difference was regulatory strategy, not tool access

High-achieving students

Goal setting
Conceptual questions
Monitoring and reflection
Iterative strategy use

Low-achieving students

Strategy selection
Procedural asking
Answer-seeking flow
Limited reflection

$t(12.07) = 4.05$ · $p < .001$ · Cohen's $d = 1.83$

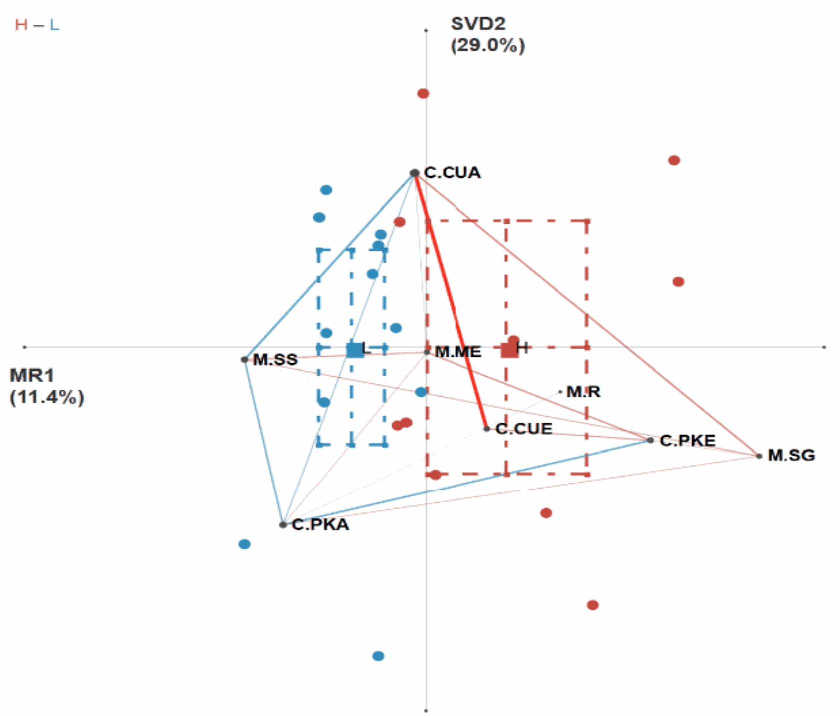
Experimental group only: high-achievers ($n = 10$) vs low-achievers ($n = 11$)

Comparison: ENA X-axis centroid difference between ability groups

ENA shows integrated versus fragmented cognitive networks

High-achievers built multidirectional cycles; low-achievers stayed procedural

Original ENA result figure from the source deck



arison of ENA network structures between high-achieving (H) and l
(L) groups

H: integrated, multidirectional regulatory network

L: fragmented, procedural, ask-execute pattern

The simplified editable schematic from v3 has been replaced by the original ENA evidence figure.

Teachers need to teach students how to regulate with AI

Availability is not enough; strategic engagement must be scaffolded

TALPer's pedagogical effectiveness cannot be assumed from tool availability alone.

Guided questioning

Model how to ask conceptual questions, not only request answers.

Structured reflection

Use prompts that make students check, explain, and revise.

Dialogue modeling

Show examples of productive TALPer-mediated dialogue.

These supports are especially critical for low-achieving students who tend toward procedural dependence.

AI feedback should invite explanation, comparison, and reflection.

Take-home: AI scaffolding depends on regulatory awareness

Thank you. Questions and discussion welcome.

An AI learning companion is not a silver bullet.

Its scaffolding effect depends on learners' regulatory awareness.

Thank you. Questions and discussion welcome.

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Future work: scaffold regulatory awareness for low-achieving students; test long-term use across math domains.

Limitations: small sample (n = 44); longer-term studies needed.